Certified Computation of Optimal Multiband Filtering Functions

Vincent Lunot, Fabien Seyfert, Stéphane Bila, and Abdallah Nasser

***Abstract*—In this paper, we focus on the problem of computing multiband filtering characteristics with a guarantee on their global optimality with respect to a Zolotarev-like criterion. An iterative algorithm based on linear programming is presented. This algorithm ensures quadratic convergence to the optimal solution. We also provide an equiripple-like criterion that allows one to check in a very simple manner whether a computed filtering function is optimal or not. The latter is used to analyze two practical design examples based on asymmetric dual-band specifications. In particular, it is shown that the selectivity of the dual-band response does not necessarily increase with the filter's order. This study yields some striking results when compared to the usual single-band situation, and introduces the idea that for certain asymmetric specifications some of the filter order values are more suitable than others. Finally, the practical implementations of the filtering devices in inline dual-mode cavities and stacked single-mode cavities are detailed.**

***Index Terms*—Differential correction algorithm, filter synthesis, multiband filter, Zolotarev problem.**

I. Introduction

IN BOTH space and terrestrial communications, advanced filtering characteristics have become a major way of improving and simplifying the architecture of systems. In particular, a dual-band filtering characteristic allows one to incorporate the two passbands within the single filter structure. The latter is an interesting replacement for the doubly multiplexed solution that combines single-band filters with a junction at the input/output. The main advantage is to save mass and volume, but also to simplify both the manufacturing and the tuning of the hardware since the multiband filter architecture can be realized with topologies and technologies commonly used for single-band filter design.

Some recent studies [1], [2] exposed methods using frequency transformations to design multiband filters. However, these lack generality. Indeed, the response is limited by symmetric specifications or by the position of the transmission zeros.

For general specifications, some optimization methods are known [3], [4]. However, they do not guarantee the optimality of the response.

In [5], an iterative algorithm based on linear programming was proposed in order to compute the "best" filtering function in the Zolotarev sense. Unlike general optimization methods that might end up at a local, but nonglobal optimum, the latter method guarantees the global optimality of the response. A drawback, however, is its poor rate of convergence, yielding a high number of necessary iterations to obtain a reasonable convergence. In this paper, modifications are introduced in the main computation loop of the algorithm that ensure quadratic convergence to the optimal solution. We also give a simple characterization of the optimal filtering function in terms of an alternation property, whereas in the single-band case, this characterization reduces to the classical equiripple property this is no longer true for the multiband situation: we give an example of an optimal, but nonequiripple response.

Two examples are constructed of the design of dual-band filters when beginning from asymmetrical frequency specifications. The ability to certify the optimality of the computed responses appears to be crucial and leads to some striking results when compared to the usual single-band situation. In the latter, the selectivity of the response increases with the filter's order. This is no longer true for dual-band specifications for which specific degrees appear to be more adapted than others. A trivial example of this is given by symmetric dual-band specifications: in this case, one can show that the optimal filtering function is always of even order. Adding an extra reflection zero to an even-order characteristic will only deteriorate the response. This type of phenomenon also occurs in a less predictable manner when dealing with asymmetric frequency specifications. To illustrate this, we analyze frequency specifications for which the best characteristic with, at most, ten reflection zeros and three transmission zeros is shown to be of 9–3 type.

Finally, we detail the practical implementation of the filtering devices respectively in aligned dual-mode cavities and stacked single-mode cavities.

II. Statement of the Synthesis Problem

A. Polynomial Structure of the Scattering Matrix

The scattering matrix associated with the classical low-pass lossless circuit prototype [6] has the following structure:

$$S = \frac{1}{E}\begin{bmatrix} F & P \\ P & (-1)^n F^* \end{bmatrix} \tag{1}$$

where n is the number of resonators. The polynomial P is of degree $m < n-1$ and satisfies the condition $P = (-1)^{n+1}P^*$ (which implies that the set of transmission zeros is symmetric with respect to the imaginary axe, i.e., paraconjugated). F is

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of degree n and monic and the denominator E is the unique Hurwitz polynomial satisfying the following spectral equation:

$$EE^* = FF^* + (-1)^{n+1}P^2. \tag{2}$$

Using the latter equations, the squared modulus of the transmission parameter is expressed as

$$|S_{21}(j\omega)|^2 = \frac{1}{1 + \left|\frac{F(j\omega)}{P(j\omega)}\right|^2} \tag{3}$$

where $D = F/P$ is known as the filtering or characteristic function.

In the case of a single passband and for given transmission zeros (i.e., P is fixed), the classical formula using the $\mathrm{Arccosh}$ function [6] allows the computation of a polynomial F that yields an equiripple quasi-elliptic filtering characteristic. The latter formula, in fact, gives the solution to the so-called third Zolotarev optimization problem that, roughly speaking, specifies in mathematical terms the notion of a "best" filtering function for a bandpass filter, whereas in the multiband situation, explicit formulas no longer exist for D, we show in the following that the original Zolotarev problem adapted to a single passband can easily be extended to take into account several passbands and stopbands.

B. General Zolotarev Problem

Given a set of passbands (see Fig. 1), i.e., a collection of intervals on the real axes $I_{1,\ldots,}I_r$ (which union we call I), as well as a collection of stopbands $J_{1,\ldots,}J_p$ (which union we call J), the "best" multiband response is such that the modulus of the transmission is as big as possible on the I intervals (passbands) and as small as possible on the J intervals (stopbands). The latter translates in a straightforward manner to the following normalized optimization problem specifying what the best filtering function is:

$$T_m^n = \left\{(F,P) \in H^n \times H^m, \left\|\frac{F}{P}\right\|_I \leq 1\right\} \tag{4}$$

$$\text{solve}: \max_{(F,P)\in T_m^n} \min_{\omega \in J} \left|\frac{F(\omega)}{P(\omega)}\right| \tag{5}$$

where H^k is the set of polynomials of degree less than k and $\|.\|_I$ is the sup norm over the set I (see Fig. 1). Indeed, if F'/P' is an optimal solution of (5) with F' monic, one may verify that setting $F(s) = F'(s/j)$ and $P(s) = j^{(n+1)}P'(s/j)/\varepsilon$ yields a scattering matrix with structure (1), with the lowest possible transmission on all the stopbands J_i, provided $|S_{21}|^2 \geq 1/(1+\varepsilon^2)$ on the passbands I_i.

C. Real Zolotarev Problem

In this study, we consider solving (5) under the additional condition that F and P are polynomials with real coefficients. This in particular implies that the synthesized scattering matrix satisfies $S_{11} = S_{22}$ and that the reflection zeros are paraconjugated, which is clearly an extra condition. On the one hand, the latter guarantees, for example, that the response can be synthesized in a cul-de-sac topology, but on the other hand, the solution to the "complex" Zolotarev problem can achieve better results (because it is less restricted).

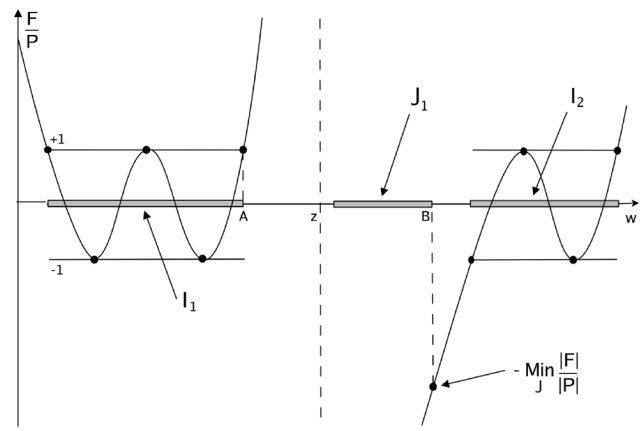

Fig. 1. Graph of function F/P with one transmission zero in z, two passbands I_1 and I_2, and one stopband J_1.

III. Linear Programming and Polynomial Approximation Problems

This section is meant as a short tutorial on the use of linear programming in connection with polynomial approximation problems like the one we just stated. Suppose we only have one stopband $J = [1.1, 2]$ and one passband $I = [-1, 1]$. We are interested in the all-pole filter of order 2 that solves the related Zolotarev problem, i.e., among all polynomials of degree less than 2 that are bounded by 1 on I, find the one with the fastest growth on J. The solution to this problem is, of course, known to be the Chebyshev polynomial $F(x) = \cos(2 \cdot \mathrm{Arcos}(x)) = 2x^2 - 1$. We will now see that this result can be recovered from a numerical algorithm in a guaranteed manner. The advantage of this procedure is that it will extend to multiband situations for which closed-form formulas are not known. Once a sign has been chosen for the polynomial $F = ax^2 + bx + c$ on J (say, positive), the original Zolotarev problem can be formulated as the following optimization problem:

$$\text{solve}: \max h$$
$$\text{subject to} \quad \begin{cases} \forall x \in J, & h \leq ax^2 + bx + c \quad \text{(i)} \\ \forall x \in I, & 1 \geq ax^2 + bx + c \quad \text{(ii)} \\ \forall x \in I, & -1 \leq ax^2 + bx + c \quad \text{(iii).} \end{cases}$$

Here, h is an auxiliary variable, which expresses the minimum of the polynomial over J. Evaluating inequalities (i) at sample points in the interval J and inequalities (ii) and (iii) at sample points in the interval I yields a set of linear inequalities in the variables (a, b, c, h). In this manner, the original Zolotarev problem is cast into a linear optimization problem under linear constraints: a linear program (LP). These types of problems have been widely studied and efficient software to solve them in a guaranteed manner exist (e.g., Cplex, MATLAB, Maple, LpSolve). Using the LP solver of MATLAB and taking 100 sample points over the intervals I and J yields the solution $a = 2.0002$, $b = 1e-12$, and $c = -1.0002$.

The advantage of this method as compared to closed-form formulas is that it generalizes to any number and any arrangement of the intervals I and J. We strongly encourage the reader to derive from what precedes a simple algorithm to solve the multiband synthesis problem in the special case of all-pole filters.

In the following sections, the general problem of filters with transmission zeros at finite frequencies is tackled. This amounts to dealing with rational fractions instead of polynomials.

The general algorithmic framework remains, however, similar and relies in particular on the use of linear programming.

IV. Sign Combinations and Characterization of the Solution

A. Sign Combinations

Our goal is now to eliminate the absolute value in (5) to get a "linear" version of the problem. If F'/P' is an optimal solution of (5) and is irreducible ($gcd(F', P') = 1$) then, as the value of the $\max\ \min$ in (5) is positive, F' has no zero in J and, as the absolute value of F'/P' is bounded by 1 over I, P' has no zero in I. Therefore, F' has a constant sign on every interval J_j and P' has a constant sign on every interval I_i. Thus there exists a sign function σ (such that $\sigma(\omega) = \pm 1$) that is constant on every interval I_i and J_j such that F'/P' has a representative in the convex set

$$A_m^n = \left\{ \begin{array}{l} (F,P) \in H^n \times H^m, \forall \omega \in J : F(\omega)\sigma(\omega) \geq 0 \\ lc(P) = 1, \left\| \dfrac{F}{P} \right\|_I \leq 1, \forall \omega \in I : P(\omega)\sigma(\omega) \geq 0 \end{array} \right\} \tag{6}$$

where $lc(P) = 1$ signifies that the coefficient of x^m in P is equal to 1.

Of course, we do not know the signs in advance, but there are only a finite number of possible combinations of them. For every combination of signs on the intervals, we therefore define a signed version of (5) by

$$\text{solve} : \max_{(F,P)\in A_m^n} \min_{\omega \in J} \frac{\sigma(\omega)F(\omega)}{|P(\omega)|}. \tag{7}$$

Solving (7) for all possible sign combinations and retaining the overall best solution yields an optimal solution of (5).

B. Characterization of the Solution

For a given sign function σ, we now give a way of testing whether a rational function of "full rank" (where no simplification between numerator and denominator occurs) is a solution of (7). The latter is based on an alternation property.

Let λ be the value of the minimum of $|F/P|$ on J. We call J^+ (respectively, J^-) the union of intervals J_i such that $\sigma(J_i) = 1$ (respectively, $\sigma(J_i) = -1$). We define the following sets of "extreme" points:

$$E^+(F,P) = \left\{ \omega \in I, \frac{F}{P}(\omega) = 1 \right\} \cup \left\{ \omega \in J^-, \frac{F}{P}(\omega) = -\lambda \right\}$$
$$E^-(F,P) = \left\{ \omega \in I, \frac{F}{P}(\omega) = -1 \right\} \cup \left\{ \omega \in J^+, \frac{F}{P}(\omega) = \lambda \right\}.$$

In Fig. 1, ten "extreme" points are plotted.

A sequence of consecutive points ($\omega_1 < \omega_2 < \cdots < \omega_k$) is called "alternant" if its points belong alternatively to the sets $E^+(F', P')$ and $E^-(F', P')$. In Fig. 1, an alternant sequence of nine consecutive points can be found (points A and B belong to the same set and cannot, therefore, appear consecutively in an alternating sequence).

"Extreme" points allow to determine whether a function is the solution of (7) or not. The following indeed holds.

- A concave maximization problem, i.e., (7), admits a unique solution.
- F'/P' is an optimal solution of "full rank" if and only if there exists a sequence of $N + 2$ frequency points $\omega_1 < \omega_2 < \cdots < \omega_{N+2}$ such that its elements belong alternatively to the sets $E^+(F', P')$ and $E^-(F', P')$ with $N = m + n$.

The latter alternant sequence is, therefore, a proof of optimality for a given filtering function.

In the single-band case, the characterization we gave is equivalent to the classical equiripple property in the passband and stopbands. However, in the multiband case, this is no longer true in general. Fig. 2(a) shows the optimal 6–4 function for the stopbands $[-2; -1.3]$, $[-0.6; 0]$, $[1.3; 2]$, and for the passbands $[-1; -0.8]$, $[0.6; 1]$. The attenuation level attained in the stopbands is of -32.2 dB, whereas the return loss is set to -20 dB. The 12 "extreme" points certify that this 6–4 nonequiriple function is the optimal solution with respect to the specifications. As pointed out by some reviewers, one might enlarge a bit the passbands and try to obtain an equiripple response with different return-loss levels in the passbands. This was done by solving the problem with following passbands $[-1; -0.75]$ and $[0.5; 1]$ and return loss levels of, respectively, -25 and -20 dB. As shown in Fig. 2(b), the optimal frequency response for these new specifications is equirriple. These new specifications are harder to meet than the preceding ones (larger passbands and higher return loss in one passband) and result in a poorer optimal attenuation level of -22.4 dB. Here again, 12 "extreme" points certify the optimality of the response.

In the following, we present a quadratic convergent algorithm for (7).

V. Algorithm

A. Geometry of the Sub-Problem

We will now study (7) from a geometric point of view. If we denote by M the value of the criterion $\min$ in (7) for a given (F, P) (M can be seen as the rejection level of F/P in the stopbands), then the convex set $E(M)$ defined by

$$E(M) = \{(F,P) \in A_m^n, \forall \omega \in J : \sigma(\omega)F(\omega) - M\,|P(\omega)| \geq 0\} \tag{8}$$

is, in a way, the set containing all the functions that have at least a rejection level M in the stopbands. Let M' be the value of the criterion $\max\ \min$ in (7) (M' is the best possible rejection). By definition of the $\max$, $E(M')$ is then the set of representatives of the optimal function F'/P'.

The key point for computing the solution of (7) is that, for $M_1 < M_2 < M' < M_3$, the following holds (see Fig. 3).

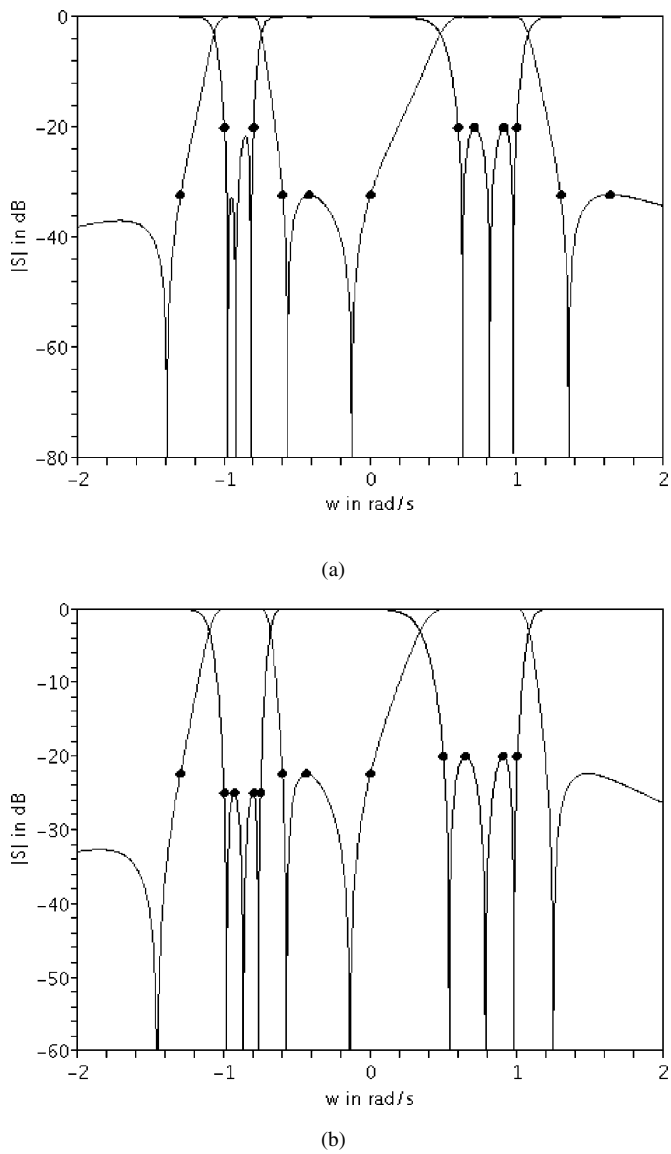

Fig. 2. (a) Optimal, but nonequiripple filtering function with six poles and four zeros. (b) Optimal 6–4 response with enlarged passbands and unequal return-loss levels in the passbands.

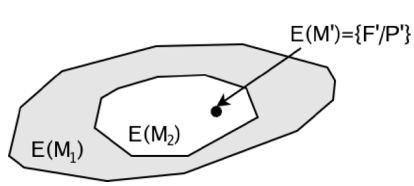

Fig. 3. Sets $E(M)$ for $M_1 < M_2 < M'$.

- $E(M_3) = \varnothing$.
- $E(M') \cong \{F'/P'\}$.
- $E(M') \subset E(M_2) \subset E(M_1)$.

Indeed, by making a hypothesis on the possible rejection level M and by checking the emptiness of $E(M)$, the following information on M' is known.

- If $E(M)$ is empty, $M' < M$.
- If $E(M)$ is nonempty, $M' \geq M$.

Therefore, a dichotomy method testing emptiness can be used to compute the optimal rational filtering function.

It is crucial to notice that the convexity of the set $E(M)$ allows to check nonemptiness using linear programming.

B. Detailed Equations for Checking Emptiness

One way of checking the emptiness of $E(M)$ is to find (F, P) in A_m^n, which maximizes the following function f_M:

$$f_M(F,P) = \min_{\omega \in J} \left(\sigma(\omega)F(\omega) - M\,|P(\omega)|\right). \tag{9}$$

Computation of (F, P) can be done by discretizing the I and J intervals. Indeed, in this way, the equations of the constraints in A_m^n become linear in the coefficients of F and P. Thus, the problem of finding (F, P) is done by solving the LP problem

$$\begin{aligned}&\text{solve}: \max h\\ &\text{subject to}\\ &\begin{cases}\sigma(y_j)F(y_j) - MP(y_j) \geq h, & \text{for all } y_j\\ \sigma(y_j)F(y_j) + MP(y_j) \geq h, & \text{for all } y_j\\ \sigma(x_j)P(x_j) \geq 0, & \text{for all } x_j\\ -\sigma(x_j)P(x_j) \leq F(x_j) \leq \sigma(x_j)P(x_j), & \text{for all } x_j\end{cases}\end{aligned} \tag{10}$$

where (x_j) [respectively, (y_j)] are a discretization of I (respectively, J).

If the maximum h is positive, then (F, P) in A_m^n, which maximizes f_M, has been computed, therefore, the set $E(M)$ is nonempty. Else, if $h \leq 0$, the set $E(M)$ is empty.

Accuracy depends, of course, on the number and placement of chosen points. On each interval, taking approximately 20 Chebyshev points gives satisfactory results.

C. Differential Correction-Like Algorithm

Instead of using dichotomy as previously suggested, we now come to an algorithm that adjusts M in a more efficient way by using the information gained from solving (10). The latter is an adaptation of the differential correction algorithm introduced by Cheney [7] for rational approximation.

Step 0: Initialization.

Choose polynomials (F_0, P_0) in A_m^n. Compute

$$M_0 = \min_{\omega \in J} \left| \frac{F_0}{P_0}(\omega) \right|. \tag{11}$$

Step k: Compute (F_k, P_k), which solves the LP problem (10) for $M = M_{k-1}$

$$f_{M_{k-1}}(F_k, P_k) = \max_{(F,P) \in A_m^n} f_{M_{k-1}}(F, P). \tag{12}$$

If $f_{M_{k-1}}(F_k, P_k) \leq 0$, return (F_{k-1}, P_{k-1}) else compute

$$M_k = \min_{\omega \in J} \left| \frac{F_k}{P_k}(\omega) \right|. \tag{13}$$

This iterative algorithm is proven to converge to the solution of (7) in the nondegenerated case (degree $P = m$). Instead of

taking the leading coefficient of P equal to 1, another normalization can be taken to ensure convergence in the general case.

For initialization, the choice of (F_0, P_0) in A_m^n does not influence the solution. One way of computing possible (F_0, P_0) is to solve (12) with M_{-1} properly chosen (e.g., $M_{-1} = 1$ generally works).

D. Improvements: Specifications and Quadratic Convergence

This algorithm can be easily generalized to compute the best filtering function subject to arbitrary specifications Ψ (Ψ is a positive function) in passbands and stopbands: in f_M, replace M by $M + \Psi(\omega)$ and in A_m^n, replace bound 1 over I by $\Psi(\omega)$.

Furthermore, the rate of convergence can be greatly increased by a slight modification of function f_M. Indeed, for a fast computation of the solution with respect to specifications Ψ, replace $f_{M_{k-1}}$ at step k by

$$f_{M_{k-1}}(F,P) = \min_{\omega \in J} \left(\frac{\sigma(\omega)F(\omega) - (\psi(\omega) + M_{k-1})\,|P(\omega)|}{|F_{k-1}(\omega)|} \right). \tag{14}$$

The associated LP problem is defined by

$$\text{solve}: \max h$$
$$\text{subject to} \begin{cases} \sigma(y_j)F(y_j) - (\psi(y_j)+M)P(y_j) \geq h\,|F_{k-1}(y_j)|, \\ \quad \text{for all } y_j \\ \sigma(y_j)F(y_j) + (\psi(y_j)+M)P(y_j) \geq h\,|F_{k-1}(y_j)|, \\ \quad \text{for all } y_j \\ \sigma(x_j)P(x_j) \geq 0, \quad \text{for all } x_j, \\ -\sigma(x_j)\psi(x_j)P(x_j) \leq F(x_j) \leq \sigma(x_j)\psi(x_j)P(x_j), \\ \quad \text{for all } x_j. \end{cases}$$

An adaptation of the proof for rational approximation (see [8]) shows that the convergence is quadratic whenever the solution is of "full rank."

VI. Examples

We consider two design examples based on asymmetric frequency specifications. Here we discuss the computation of the filtering functions, whereas their practical implementations are considered in Section VI-A.

Example 1

A first example is taken from [5] with the following electrical specifications:

- return loss at 20 dB in the passbands ($I_1 = [-1, -0.625]$ and $I_2 = [0.25, 1]$ on the ω-axis, normalized frequency);
- rejection at 15 dB in the lower and upper stopbands ($J_1 =]-\infty, -1.188]$ and $J_3 = [1.212, +\infty[$) and 30 dB in the intermediary stopband ($J_2 = [-0.5, 0.125]$).

One may first think of computing a 10–3 filtering characteristic to fit in the latter specifications. Using the previously presented algorithm and working for practical reasons with finite intervals (so the two chosen "outside" stopbands are set to $[-10, -1.188]$ and [1.212, 10]), we obtain the filtering function plotted in Fig. 4. Only nine transmission zeroes and 14 "extreme" points appear on the graph, which seems at first glance to contradict the theory or to indicate that something is wrong with our numerical implementation. A closer inspection of the obtained function indicates, however, that the lacking "extreme" point is situated in the left limit of the first stopband, i.e., in $\omega = -10$, together with a reflection zero that was rejected at $\omega = -100$. If we increase the size of the left stopband, the reflection zero is rejected further towards infinity. This amounts to saying that the optimal characteristic with, at most, ten reflection zeros (respectively, at most, three transmission zeros) is, in fact, of 9-3 type. In some sense, the optimization process indicates that there is no way to improve this 9-3 filtering function by adding an extra reflection zero. Note that here the ability to certify the optimality of the computed filtering function is crucial. Someone using a generic optimizer may insist on finding a better starting point for his optimization process or try by all means to restrict the location of reflection zeros: by the optimality argument, this can only yield a poorer result.

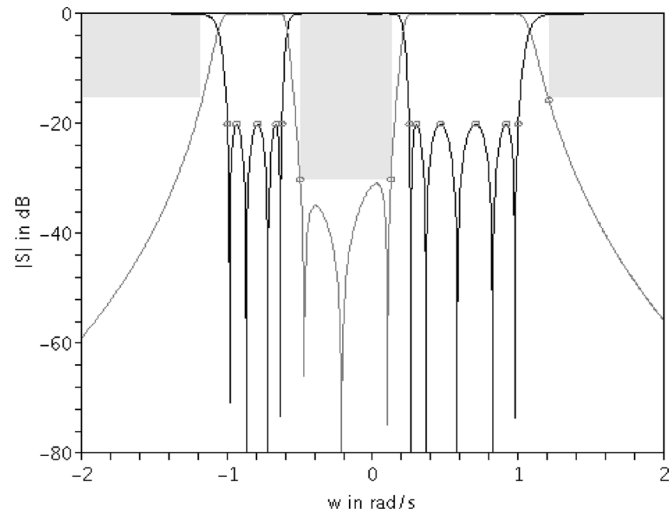

Fig. 4. Optimal transmission and reflection parameters (example 1).

Example 2

A second example from [9] is taken whose electrical specifications are defined by the following:

- return loss at 23 dB in the passbands ($I_1 = [-1, -0.383]$ and $I_2 = [0.383, 1]$ on the ω-axis);
- in the lower stopband ($J_1 =]-\infty, -1.864]$), rejection is set at 10 dB on $]-\infty, -1.987]$ and 15 dB on $[-1{,}987, -1{,}864]$. Rejection is set at 20 dB in the intermediary stopband ($J_2 = [-0.037, -0.012]$) and 40 dB in the upper stopband ($J_3 = [1.185, +\infty[$).

Here again, one may think of using an 8–3 characteristic for a realization in extended box topology [11]. However, the same phenomenon as in example 1 occurs, and the optimal solution appears to be of type 7–3. The slight difference between the filtering function in Fig. 5 and the one in [9] is due to the fact that reflection zeros were originally laboriously optimized "by hand." Here, transmission and reflection zeros are optimized simultaneously. In the upper stopband, the rejection level is lower than in [9], but is improved in the other stopbands and, contrary

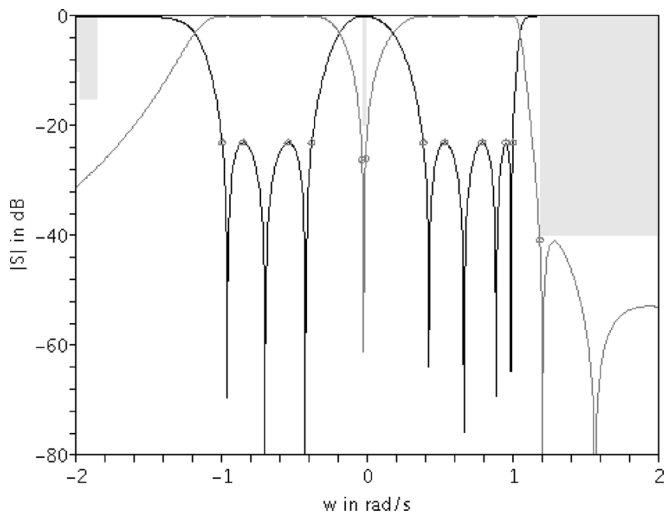

Fig. 5. Optimal transmission and reflection parameters (example 2).

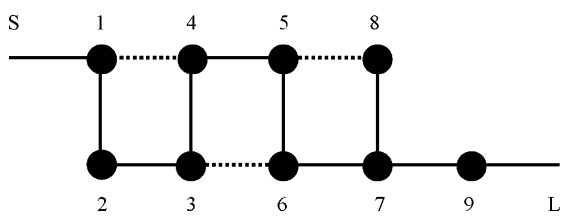

Fig. 6. Extended-box coupled resonator network for the realization of the ideal 9–3 dual-band response in Fig. 4.

to the original solution, the return loss is equiripple in the two passbands.

VII. Design and Implementation at Microwave Frequencies

Example 1: Nine-Pole Three-Zero Dual-Band Filter Implemented in Inline Dual-Mode Cavities

The low-pass specifications given in Fig. 4 correspond to the following passbands and stopbands at microwave frequencies: the two passbands are, respectively, $I_1 = [8.28, 8.31]$ GHz and $I_2 = [8.38, 8.44]$ GHz and the three stopbands are, respectively, $J_1 = [0, 8.265]$ GHz, $J_2 = [8.32, 8.37]$ GHz, and $J_3 = [8.457, +\infty[$ GHz.

From these ideal parameters, a coupled resonator network has to be derived for realizing the desired number of transmission and reflection zeros. The network is chosen to be an extended-box one (see Fig. 6) since this topology allows a practical implementation of the filtering function with aligned dual-mode cavities. The technology selected for realizing the microwave filter consists in cylindrical cavities working on their dual-mode TE_{111} and coupled by rectangular irises, as shown in Fig. 7.

Applying an exhaustive coupling matrix synthesis [11], 22 real solutions have been found to realize the optimal function with the extended-box network.

A particular solution is then selected and a computer-aided design (CAD) model is tuned, applying a coupling matrix identification at each tuning step [10]. However, in this case, an exhaustive computation of all the solutions to the coupling matrix synthesis problem is necessary for recognizing the solution to be tuned. In case of ambiguity between several identified solutions, the solution that corresponds to the CAD model can be recognized by perturbing some coupling elements (dimensions of irises or screws) and by studying the coherency on the solution modifications (corresponding coupling values). The CAD model is a finite-element model. Metallic losses are not considered during CAD tuning for facilitating comparison with the synthesized lossless rational function. Moreover, no particular action, i.e., predistortion, is done for compensating losses in the current synthesis.

A hardware prototype of the filter has been built with brass. The unloaded quality factor is approximately 4000, but can be improved using silver-plated cavities. However, measured and simulated results are in good agreement, as shown in Fig. 8. Insertion losses are 2.15 dB in the first passband and 1.45 dB in the second one.

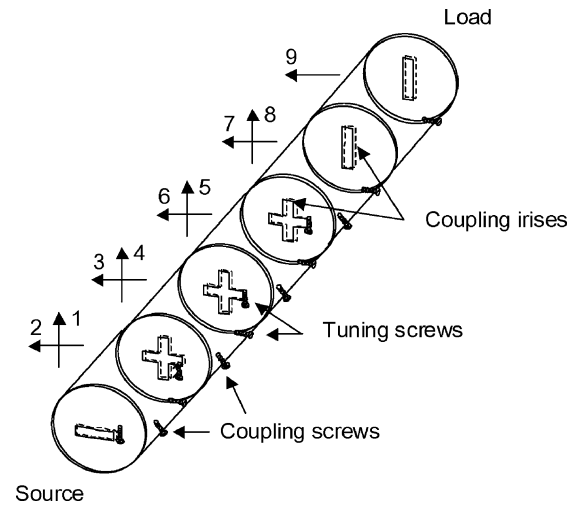

Fig. 7. Implementation of the nine-pole three-zero dual-band filter with inline dual-mode cylindrical cavities, network topology illustrated in Fig. 6.

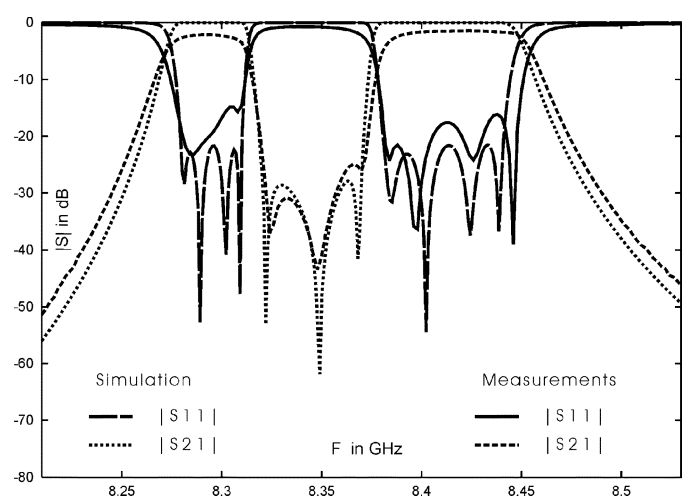

Fig. 8. Measurements and simulation of the nine-pole three-zero dual-band filter physically illustrated in Fig. 7.

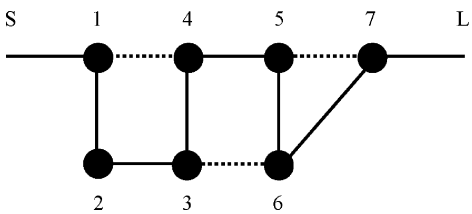

Fig. 9. Pseudo extended-box coupled resonator network for the realization of the ideal 7–3 dual-band response in Fig. 5.

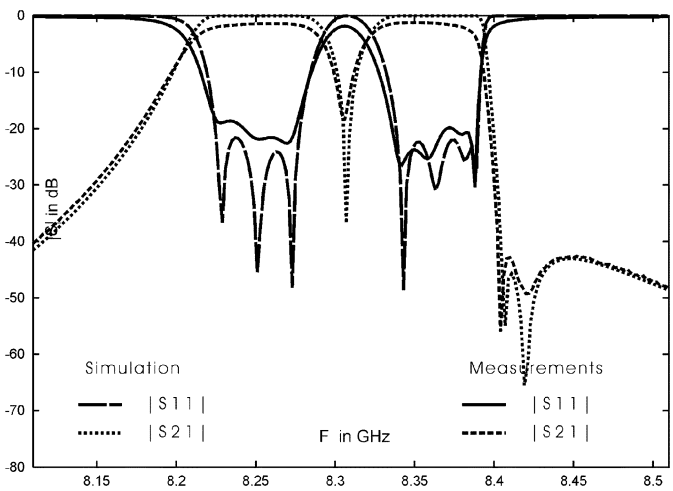

Fig. 10. Measurements and simulation of the 7–3 dual-band filter, network topology illustrated in Fig. 9.

Example 2: Seven-Pole Three-Zero Filter Implemented in Stacked Single-Mode Cavities

At microwave frequencies, the low-pass specifications shown in Fig. 5 match into two passbands, respectively, at $I_1 = [8.228, 8.278]$ GHz and $I_2 = [8.34, 8.39]$ GHz, and three stopbands, respectively, at $J_1 =]-\infty, 8.158]$ GHz, $J_2 = [8.306, 8.308]$ GHz, and $J_3 = [8.405, +\infty[$ GHz.

The coupled-resonator network, which is selected for realizing the latter filtering function, is the pseudo extended-box topology presented in Fig. 9.

This configuration of the coupled-resonator network leads to three real solutions for realizing the ideal filtering characteristic.

A solution is chosen for being implemented in stacked single-mode rectangular cavities, as described in [9]. The CAD model and the practical hardware are tuned using an exhaustive coupling matrix identification. Measurement results of the brass-made prototype are compared with simulations in Fig. 10. Insertion losses are, respectively, 1.4 and 1.25 dB in the passbands.

VIII. Conclusion and Perspectives

We have presented an iterative algorithm with a quadratic convergence rate to perform the optimal synthesis of multiband filtering functions. We also provided a simple equiripple-like optimality criterion that allows one to check for the optimality of any filtering function. The main advantages of this approach, as opposed to direct optimization schemes, are to provide the user with some optimal placements for the reflection and transmission zeros while validating the result. Based on the latter, we showed that, for some dual-band specifications, an increase of the filter's order does not necessarily yield an improvement of its selectivity: this is a major difference as compared to the usual single-band situation and, to the best of our knowledge, the analysis of this phenomenon is new to the filtering community. Finally, the process was validated by the design of two asymmetric dual-band bandpass filters.

References

[1] R. J. Cameron, M. Yu, and Y. Wang, "Direct-coupled microwave filters with single and dual stopbands," *IEEE Trans. Microw. Theory Tech.*, vol. 53, no. 11, pp. 3288–3297, Nov. 2005.

[2] G. Macchiarella and S. Tamiazzo, "Design techniques for dual-passband filters," *IEEE Trans. Microw. Theory Tech.*, vol. 53, no. 11, pp. 3265–3271, Nov. 2005.

[3] S. Amari, "Synthesis of cross-coupled resonator filters using an analytical gradient-based optimization technique," *IEEE Trans. Microw. Theory Tech.*, vol. 48, no. 9, pp. 1559–1563, Sep. 2000.

[4] M. Mokhtaari, J. Bornemann, K. Rambabu, and S. Amari, "Coupling-matrix design of dual and triple passband filters," *IEEE Trans. Microw. Theory Tech.*, vol. 54, no. 11, pp. 3940–3946, Nov. 2006.

[5] V. Lunot, S. Bila, and F. Seyfert, "Optimal synthesis for multi-band microwave filters," in *IEEE MTT-S Int. Microw. Symp. Dig.*, Jun. 2007, pp. 115–118.

[6] R. J. Cameron, "General coupling matrix synthesis methods for Chebyshev filtering functions," *IEEE Trans. Microw. Theory Tech.*, vol. 47, no. 4, pp. 433–442, Apr. 1999.

[7] E. W. Cheney, *Approximation Theory*. New York: Chelsea Press, 1982.

[8] D. Braess, *Nonlinear Approximation Theory*. Berlin, Germany: Springer-Verlag, 1986.

[9] S. Bila, R. J. Cameron, P. Lenoir, V. Lunot, and F. Seyfert, "Chebyshev synthesis for multi-band microwave filters," in *IEEE MTT-S Int. Microw. Symp. Dig.*, Jun. 2006, pp. 1221–1224.

[10] S. Bila, D. Baillargeat, S. Verdeyme, M. Aubourg, P. Guillon, F. Seyfert, J. Grimm, L. Baratchart, C. Zanchi, and J. Sombrin, "Direct electromagnetic optimization of microwave filters," *IEEE Micro*, vol. 2, no. 1, pp. 46–51, Mar. 2001.

[11] R. J. Cameron, J. C. Faugère, and F. Seyfert, "Coupling matrix synthesis for a new class of microwave filter configuration," in *IEEE MTT-S Int. Microw. Symp. Dig.*, Jun. 2005, pp. 119–122.

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